

Exercise 1

Part 1 – Effects of automated Tests:

1. **Recap: What is the difference between functional and non-functional requirements?:** a functional requirement describes the behavior of the system what it should do, it describes features. Non-functional requirements describe how efficient, maintainable, robust, secure, reliable our system is.
2. **What is the relationship between automated tests and functional requirements?:** automated tests verify the functional requirements by checking the system response according to the requirements we have written, since we specified what should happened on action when writing the requirements.
3. **What is the relationship between automated tests and non-functional requirements?:** automated tests checks whether the system responses quickly, is secure, handles data loads, and traffic loads with grace, these are non-functional requirements that automated tests can verify. Tho not all non-functional requirements are verifiable through automated tests, some needs to be done manually.
4. **How do automated tests help with collaboration between software engineers and non-technical development roles (such as product owners)?:** Automated tests provide measurable results to verify the requirements written by non-technical team members, creating a shared understanding of what has been implemented. They also support continuous integration, allowing everyone to see the system's current behavior and status in real time.
5. **How do automated tests help with collaboration in-between software engineers?:** automated tests help with continuous integration, meaning that the entire team can see the same results and failures, this helps faster understanding and bug detection. Automated tests also provide verification for functional requirements.

Part 2 – Code Example:

These answers are based on the code in the file called temp code.txt

1. **What functional requirements can you glean from this code?:** The system shall store the user's name and balance amount. The system shall show the user the balance and the owner name. the system shall let the user withdraw money when they hit the withdraw button, then subtract it from their balance. If the user wants to deposit money the system shall let them and then increase the balance.
2. **What non-functional requirements can you glean from this code?:** the system shall throw an error when the user enters a negative number when depositing into the account. The system shall throws an error when the user tries to withdraw more money than they have. The system shall throws an error when the user tries to withdraw negative amount of money.

3. **How could these specific automated tests help you to collaborate with another technical or non-technical stakeholder?:** From a technical perspective, automated tests help developers quickly identify which part of the code is broken or needs fixing. From a non-technical perspective, they provide a clear, measurable way to verify that requirements are being met, creating a shared understanding between engineers and stakeholders.
4. **Can you see a common structure in the automated tests? If yes: What is it and what is the point of such a structure?:** they all use the assertion API, they all compare the result to what is expected, they all have meaningful method name. they all have annotation test above them. provide consistent, repeatable verification of system behavior.

Part 3 – Revisiting the Lecture:

1. **Explain why testing is important using two examples not from the lecture. Find two known errors and explain how these could have been prevented using testing:** Testing is important because it helps catch errors before they reach users, preventing financial losses or safety risks. the Knight Capital glitch in 2012 caused a trading algorithm to behave incorrectly, which could have been prevented with automated tests simulating edge cases. Toyota unintended acceleration in 2009, caused by faulty throttle control software, which could have been avoided with unit and integration testing
2. **Explain in your own words: How do Quality Assurance (QA) and Testing relate to one another:** Quality assurance is a framework that defines the practices and requirements that we need to build quality into our software, the tests are there to verify that the framework was followed by providing measurable results.
3. **Collaboration: What tools did you use for collaboration / documentation in the last two semesters? Which features did you perceive es useful for collaborating?:** github, gitlab CI/CD, Jira. We used github to store out technical documentation so when we need to expand the implementation we know what tools, and practices to follow. We used gitlab pipeline to see the same test results and to fix bugs faster. We used jira to pull tasks, definition of done.